

The Rational Use of Antibiotics and prescription writing

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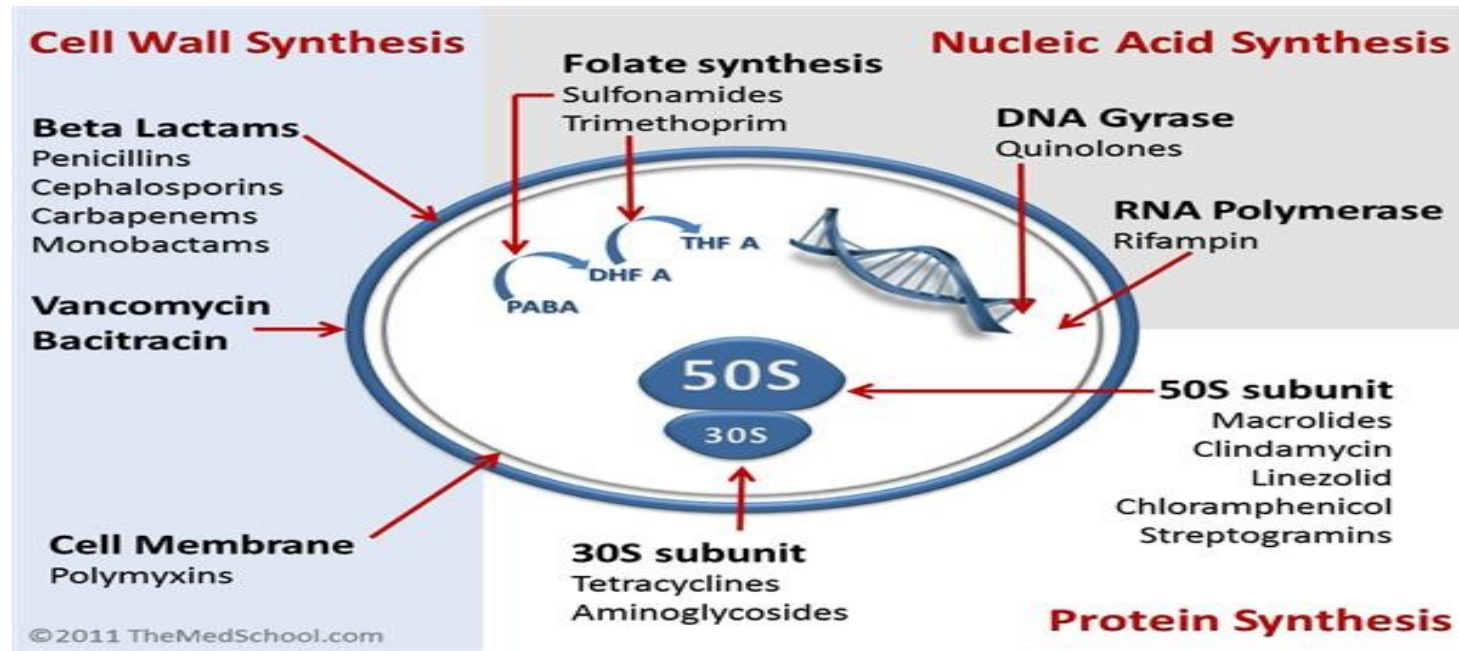
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INTRODUCTION

- Antibiotic: a substances produced by microorganisms that selectively inhibit the growth of or kill other microorganisms at very low concentration.
- Antimicrobial: a broader term for any agent that kills or inhibits microorganisms (bacteria, viruses, fungi and parasites).

Basic Pharmacology

- Pharmacokinetic: describes what the body does to the drug.
- Pharmacodynamics : describes what the drug does to the bacteria.



Classificaton of drugs

☐ **Based on effect on the bacteria:**

- I. Bactericidal.
- II. Bacteriostatic.

☐ **Based on pharmacodynamic activity:**

- I. Time dependent.
- II. Concentration dependent.
- III. Exposure dependent.

Principles of Antibacterial Therapy

Antibacterial therapy is the appropriate use of antibiotics to prevent or treat bacterial infections while maximizing clinical cure and minimizing toxicity, resistance , unnecessary costs

1. Empiric therapy.
2. Definitive (Targeted) therapy.
3. Prophylactic therapy.

1. Empiric Therapy

Therapy is considered empirical when the causative agent has yet to be determined and therapeutic decisions are based on

- The severity of illness,
- The likely pathogens and
- Prior therapy, and
- Relevant epidemiologic factors.

When is Empiric Therapy Used?

- Life-threatening infections (e.g., sepsis, meningitis, pneumonia, complicated UTI).
- Severe clinical deterioration.
- When diagnosis is likely based on history, physical exam, and epidemiology.

Key principles of empiric therapy

Principle

- **Identify likely pathogens**
- **Use local antibiogram**
- **Choose appropriate spectrum**
- **Adjust for host factors**
- **Start after obtaining samples**

Explanation

Based on infection site, clinical setting, and patient risk factors

Choose agents based on **local resistance** data. Resistance patterns vary by hospital/region.

Use **broad-spectrum** agents if needed initially. Narrow down after diagnostic results.

Consider age, renal/liver function, allergies, immune status, pregnancy, drug interactions.

Always collect blood, urine, sputum, or CSF before starting empiric antibiotics.

Indication for Antibiotic use

- A. Proven bacterial infection.
- B. Strongly suspected bacterial infection (empiric therapy).
- C. Surgical prophylaxis.
- D. Medical prophylaxis (selected high risk patients).

❖ Factors Affecting Antibiotics Choice

- A. Patient factor.
- B. Infection factor.
- C. Drug factors.

2. Definitive therapy

- Is antibiotic therapy adjusted after the causative organism and its susceptibility are identified by culture.
- Usually after 48-72 hrs after culture are obtained.

Principles include

- ✓ Review culture results.
- ✓ Identify the pathogen.
- ✓ Select the narrowest effective antibiotic.
- ✓ Adjust dose if needed.
- ✓ De-escalate from broad –spectrum therapy whenever appropriate.
- ✓ Continue treatment for the recommended duration.

3. Prophylactic Therapy

- Is the administration of antibiotics to prevent infection in patients at increased risk.

A. Surgical prophylaxis : to prevent surgical site infection.

- Give the antibiotic with in 60 minute before the surgical incision and 120 minutes for antibiotics requiring longer infusion.

B. Medical prophylaxis

- used in selected high risk patients e.g Rheumatic fever prophylaxis (benzathine penicillin), meningococcal exposure (rifampicin, ciprofloxacin, or ceftriaxone) , endocarditis prophylaxis.

Culture and Sensitivity Testing

- Definition

- ❑ **Culture:** a laboratory process of growing microorganisms from a clinical specimen to identify the causative pathogen.

- ❑ **Sensitivity (susceptibility):** laboratory testing that determines which antibiotics are effective against the isolated organism.

Purpose

- ❖ Identify the causative pathogen.
- ❖ Guide appropriate antibiotic selection.
- ❖ Enable de-escalation from broad to narrow spectrum antibiotics.
- ❖ Reduce unnecessary broad-spectrum antibiotic use.
- ❖ Prevent antimicrobial resistance.
- ❖ Reduce costs.
- ❖ Improve patient outcome.

Interpretation

- After the organism is identified, the laboratory reports its susceptibility to different antibiotics.

The report classifies antibiotics as

- **S- Susceptible:** the organism is likely to respond to antibiotic at standard doses.
- **I- Intermediate:** antibiotic may work if higher doses or optimized doses are used.
- **R-Resistant :**the antibiotic is unlikely to be effective and should not be used.

Minimum Inhibitory Concentration (MIC)

- The lowest concentration of an antibiotic that inhibits visible bacterial growth.
- Helps determine how susceptible a bacterium is to an antibiotic.
- Lower MIC values generally indicates greater susceptibility.

Antibiotic resistance

- Antimicrobial resistance is the ability of microorganisms to survive and/or multiply in the presence of tolerable doses of antimicrobial medicines.
- Antimicrobial resistance can be:
 - Natural.
 - Acquired.

Mechanisms for antimicrobial resistance

1. Alteration of drug-binding sites.
2. Production of enzymes inactivate drug.
3. Synthesis of compound antagonize drug actions.
4. Efflux pumps.
5. Decreased permeability of bacterial cell wall.

Types of antimicrobial resistance

There are different types of antimicrobial resistance.

These include:

- **Multi Medicine Resistance (MDR)** (resistance to two or more medicines).
- **Cross-resistance**, (resistance that occurs among two different antimicrobials).
- **Co-resistance**, (more than one mechanism of resistance is involved by the same organism for a given antimicrobial).

The Rational use of antibiotics

- According to WHO:-

"Patients receive medications appropriate to their clinical needs, in doses that meet their own individual requirements, for an adequate period of time, and at the lowest cost to them and their community.

Principles of Rational Antibiotics use

- ✓ Use antibiotics only for suspected or proven bacterial infections.
- ✓ Obtain appropriate specimens for culture before starting antibiotics when ever possible.
- ✓ Begin empiric therapy based on the likely pathogen, age, site of infection, and local resistance pattern.
- ✓ Modify (de-escalate) treatment once culture and susceptibility results are available.

...

- ✓ Choose the narrowest-spectrum antibiotic that effectively treats the infection.
- ✓ Use the correct dose, route , and duration.
- ✓ Monitor clinical response and adverse effects.
- ✓ Avoid unnecessary antibiotic use to reduce antimicrobial resistance.

BE AWARE
OF ANTIOTBIOTIC RESISTANCE

A Guide to the Classification and Monitoring of Antibiotics Use



ACCESS

- Affordable
- Widely available
- For common infections
- Lower risk of promoting AMR
- Low risk of toxicity

WATCH

- Higher risk of resistance
- Reserved for severe infections
- Higher risk of toxicity
- Key targets for monitoring



RESERVE

- Last-resort options
- The highest resistance potential
- For multi-drug-resistant infections

Based on WHO AWaRe classification of antibiotics for evaluation and monitoring of use, 2023

S Glaschütz

Adapted WHO EML AWaRe List for England



Improvement

Access	Watch	Reserve
Amoxicillin / ampicillin	Amikacin, tobramycin, etc	Aztreonam
Penicillin – all forms	Macrolides	Ceftobiprole,
Co-trimoxazole	Most cephalosporins	Ceftaroline
Doxycycline	Chloramphenicol	Ceftazidime-avibactam
Flucloxacillin	Fluoroquinolones	Ceftolozane-tazobactam
Fosfomycin oral	Clindamycin	Colistin
Fusidate	Co-amoxiclav	Daptomycin
Gentamicin	Other tetracyclines	Carbapenems
Metronidazole	Fidaxomicin	Fosfomycin IV
Nitrofurantoin	Piperacillin-tazobactam, etc	Linezolid / tedizolid
Pivmecillinam	Temocillin	Televancin
Tetracycline	Vancomycin, teicoplanin	Tigecycline
Trimethoprim		

Stewardship program

- Is an organized antimicrobial management program that can be undertaken to improve antimicrobial usage.

Goal of antibiotic stewardship programs

1. Reduce antibiotics consumption and inappropriate use.
2. Improve patient outcomes and decrease morbidity and mortality.
3. Increase adherence/utilization of treatment guidelines.
4. Reduce adverse drug events.
5. Decrease or limit antibiotic resistance.
6. Reduce healthcare cost.

5D'S of antibiotic stewardship

- ✓ Right DRUG.
- ✓ Correct DOSE.
- ✓ Right DRUG ROUTE.
- ✓ Suitable DURATION.
- ✓ Timely DE-ESCALATION.

De-escalation

- Antimicrobial de-escalation is a primary antimicrobial tool to prevent the emergence of drug-resistant bacteria.
- The term de-escalation describe the practice of using culture results as a basis for switching from broad-spectrum to more narrow spectrum.
- It may also include changing administration from the IV to PO, or discontinuing antimicrobials if infection has been ruled out.
- Key Takeaway: Antimicrobial de-escalation is appropriate to consider in any clinically improving patient after 2-4 days of empiric therapy.

Management of sepsis.

❑ **Non pharmacologic**

- Maintenance of body temperature (Kangaroo mother care, radiant warmer, incubator).
- Adequate calorie and fluid maintenance.
- Correction of associated metabolic abnormalities.

Table 148.10 Management and Prevention of Neonatal Sepsis

CONDITION	THERAPY	ADDITIONAL CONSIDERATIONS
EMPIRICAL MANAGEMENT		
Early-onset sepsis	Ampicillin + aminoglycoside 10 days for bacteremia; 14 days for GBS and uncomplicated meningitis; extend to 21-28 days for complicated infections	• Consider a third-generation cephalosporin or carbapenem for meningitis or severe illness with high suspicion for gram-negative infection • Tailor therapy to pathogen • Should discontinue empiric therapy if pathogen not isolated
Late-onset sepsis	Vancomycin or antistaphylococcal penicillin (i.e., nafcillin, oxacillin) + aminoglycoside Duration dependent on pathogen and site	• Gram-positive coverage should be based on local epidemiology, MRSA colonization, CONS risk factors, and clinical presentation • Aminoglycoside-based regimen preferred to cephalosporin given reduced risk of resistance • Consider cephalosporin if meningitis suspected • Consider a carbapenem if third-generation cephalosporin recently received • Consider amphotericin for fungal etiologies • Tailor therapy to pathogen • Consider discontinuation of therapy if pathogen not isolated
NONANTIMICROBIAL TREATMENT STRATEGIES		
Recombinant G-CSF	Enhance neutrophil number and function, but	Insufficient evidence to support the clinical use of G-CSF
Recombinant GM-CSF	no reduction in infection when administered as	or GM-CSF either as treatment or prophylaxis to prevent

Pharmacologic

Till the culture report is collected start with broad-spectrum antibiotics, which includes

- Penicillin and Aminoglycoside.

First line

- Ampicillin, 100mg /kg/day every 12 hours IV.for 10 days.

PLUS

- Gentamicin, 5mg/kg /day IV Daily for 10 days

Alternative

- Penicillin G Sodium Crystalline, 50,000IU/kg QID for ten days

PLUS

- Gentamicin, 5mg/kg /day IM Daily for 10 days.

Management of acute otitis media.

❑ Non Pharmacologic

- Drink lots of fluid.
- Paracentesis when indicated.

❑ Pharmacologic

First line: Amoxicillin

- Children under 6 years; 125mg/5ml, 250mg/5ml 1 tsp. P.O., TID for ten days
- Children above 6 years; 250 mg P.O., TID for ten days.

Alternatives

- **Ampicillin**, 100-200mg/kg IV into divided doses QID for children for 10 days.

- Amoxicillin/Clavulanate, 312mg/5ml suspension P.O. TID for 10 days OR 156mg/5ml suspension P.O. TID for 10 days for children.
- Ciprofloxacin, 500mg P.O., BID FOR 7-10 days.
- Ciprofloxacin ear drop eye ear solution, 0.3%, 5ml. 2 – 3 drops twice daily.
- Chloramphenicol, Solution ear drop, 1%, 2%, 5%. Apply 2–3 drops 2–3times daily.
- Paracetamol, 30-40mg/kg/24hrs.

Management of acute pharyngitis

Table 430.4 Recommended Treatment for Acute Streptococcal Pharyngitis

MOST PATIENTS				
	WEIGHT <27 KG	WEIGHT ≥27 KG	ROUTE	DURATION
Amoxicillin	50 mg/kg once daily (maximum 1,000 mg)		Oral	10 days
Penicillin V	250 mg bid	500 mg bid	Oral	10 days
Benzathine penicillin G	600,000 units	1.2 million units	IM	Once
Benzathine penicillin G + procaine penicillin G	900,000 units + 300,000 units	900,000 units + 300,000 units	IM	Once
PENICILLIN-ALLERGIC PATIENTS				
	ORAL DOSE	FREQUENCY	DURATION	
Cephalosporins*	Varies with agent chosen		10 days	
Erythromycins				
Ethylsuccinate	40 mg/kg/day up to 1,000 mg/day	bid	10 days	
Estolate	20-40 mg/kg/day up to 1,000 mg/day	bid	10 days	
Clarithromycin	15 mg/kg/day up to 500 mg/day	bid	10 days	
Azithromycin†	12 mg/kg day 1; 6 mg/kg days 2-5	qd	5 days	
Clindamycin	20 mg/kg/day up to 1.8 g/day	tid	10 days	

Management of pneumonia

Pharmacologic

1. Antipyretic

First line

- Paracetamol, 10-15mg /kg P.O., up to 4 times a day for the relief of high fever equal to or above 39 °C.

Alternatives

- Ibuprofen, 5 – 10mg/kg/dose every 6 – 8hr P.O. (max. 40mg/kg/24hours).

2. Antibiotic

First-line

- Amoxicillin, 30-50mg/kg/24hourss P.O. TID for 7 days.

Alternatives

- Azithromycin, 10mg/kg/24 P.O., once daily for 3 days.

Severe Pneumonia

- Benzyl penicillin, 50,000units/kg/24hours IM OR IV QID for at least 3 days.
- N.B. When the child improves switch to oral Amoxicillin: 30-50mg/kg/24 hours 3 times a day. The total course of treatment is 10 days.
- If the child doesn't improve within 48 hours, switch to ceftriaxone 50mg/Kg/24 hours IM/IV for 5 days.

Presumed Atypical pneumonia

- Azithromycin, 10mg/kg P.O., on day 1, 5mg/kg on day 2-5.

Management of UTI

Supportive care

- The child should be encouraged to drink or breastfeed regularly in order to maintain a good fluid intake, which will assist in clearing the infection and prevent dehydration.
- If the child has pain, treat with paracetamol; avoid non-steroidal anti-inflammatory drugs (NSAIDs).

Pharmacologic

- Treat the child as an outpatient.
- Give an oral antibiotic for 7–10 days, except:
 - ✓ When there is high fever and systemic upset (such as vomiting or inability to drink or breastfeed).
 - ✓ When there are signs of pyelonephritis (loin pain or tenderness).
 - ✓ For infants.

- ✓ Give oral co-trimoxazole (10 mg/kg trimethoprim and 40 mg/kg sulfamethoxazole every 12 h) for 5 days.
- ✓ If there is a poor response to the first-line antibiotic, give gentamicin (7.5 mg/kg IM or IV once daily) plus ampicillin (50 mg/kg IM or IV every 6 h) or parenteral cephalosporin.
- ❑ Consider complications such as pyelonephritis or septicaemia.
- ✓ Treat young infants aged < 2 months with gentamicin at 7.5 mg/kg IM or IV once daily until the fever has subsided; then review, look for signs of systemic infection, and, if absent, continue with oral treatment.

Management of diarrhea

- ❑ Fluid deficit is the amount of fluid lost from body due to dehydration.
- Fluid deficit = weight × DHN% × 10.
- ❑ Rehydration therapy is giving fluids to correct the existing fluid deficit.
- ❑ Maintenance therapy is a fluid given to meet normal daily water and electrolyte requirements when the child cant drink enough water.
- ❑ Replacement therapy means replacing ongoing loss that continue after initial rehydration has been completed.

❑ **Non pharmacologic**

Plan A (If no dehydration, treat diarrhoea at home).

- Treat the child as an outpatient.
- Counsel the mother on the four rules of home treatment:
 - ✓ Give extra fluid.
 - ✓ Give zinc supplements.
 - ✓ Continue feeding.
 - ✓ Return to the clinic if:

the child becomes sicker, is unable to drink or breastfeed, drinks poorly, develops a fever or has blood in the stool.

- ❖ If the child shows none of these signs but is still not improving, advise the mother to return for follow-up after 5 days.

- ✓ Show the mother how much fluid to give in addition to the usual fluid intake:
 - o Up to two years-50 to 100 ml after each loose stool.
 - o Two years or more:100 to 200 ml after each loose stool.

- ✓ If the child vomits, wait 10 minutes, then continue but more slowly. Continue giving extra fluid until the diarrhoea stops.

Dangerous fluids not to be given

- ❖ Drinks sweetened with sugar.
- ❖ Commercial carbonated beverages.
- ❖ Commercial fruit juices.
- ❖ Sweetened tea.
- ❖ Fluids with stimulant, diuretic or purgatives effects (e.g., coffee).
- ❖ Some medicinal teas or infusions.

Plan B (Treat some dehydration with ORS in Clinic)

- Replacement fluid with ORS 75ml/kg/4hr then re assess for response and reclassify.
- For infants less than 6 months who are not breastfed, also give 100-200 ml clean water during the 4-hour period.
- Continue breastfeeding whenever the child wants.

- ❖ Amount of ORS to be given during the first 4 hours depending on the age of the child.

Age	Up to 4 Months	4 Months up to 12 months	12 months up to 2 years	2 years up to 5 years
Weight	6 kg	6-10 kg	10-12 kg	12-19 kg
ORS in ml	200-400	400-700	700-900	900-1400

Severe dehydration (Plan C)

- Requires treatment with IV fluids (RL or NS).

AGE	First give 30 ml/kg in:	Then give 70ml/kg in:
Infants (Under 12 months)	1 hour*	5 hours
Children 12 months up to 5 years)	30 minutes*	2 ½ hours

- Repeat once if radial pulse is still very weak or not detectable.
- Reassess the child every 1-2 hours. If hydration status is not improving, give the IV drip more rapidly.
- Also give ORS (about 5 ml/kg/hour) as soon as the child can drink: usually after 3-4 hours (infants) or 1-2 hours (children).
- Reassess an infant after 6 hours and a child after 3 hours and classify dehydration.

- Suspect cholera in children > 2 years old who have acute watery diarrhoea and signs of severe dehydration or shock, if cholera is present in the area.
- Give an oral antibiotic to which strains of *V. cholerae* in the area are known to be sensitive.
- Possible choices are: erythromycin, ciprofloxacin and co-trimoxazole.
- Prescribe zinc supplementation as soon as vomiting stops.

❑ Pharmacologic

Vitamin A,

- 50,000 IU for age < 6 months,
- 100,000 IU for age 6 – 12 months,
- 200,000 IU for those > 12 months.
- ❖ **ADRs:** diarrhoea, vomiting, irritability, drowsiness.
- ❖ **C/Is:** renal impairment.
- ❖ **D/Is:** cholestyramine or colestipol reduces its absorption.

Zinc supplementation

Age $\leq$ 6 months: half tablet (10 mg) per day for 10–14 days.

Age $\geq$ 6 months: one tablet (20 mg) per day for 10–14 days.

- ❖ It helps the child's recovery, reduces the duration and severity of the episode, and lowers the incidence of diarrhoea in the following 2–3 months.

Management of impetigo

❑ Non pharmacologic

- Careful removal of crusts by bathing with Normal saline or Hydrogen peroxide ensures more rapid healing.

❑ Pharmacologic

Topical: for localized

First line

- Mupirocin, applied thin film of 2% cream/ointment 2-3 times a day for 10 days.

❖ **ADRs:** irritaion.

Alternatives

- Fucidic acid, applied thin film of 2% cream 2-3 times in a day for 10 days.

❖ **ADRs:** irritation.

N.B. Anti-bacterial ointments are usually applied after the wet lesion has dried.

Systemic: for extensive

First line

- Cloxacillin.

Alternative

- Cephalexin, 250mg to 500mg P.O., QID for 7 to 10 days for adults; 25 to 100mg/kg P.O. in 4 divided doses for 7 to 10 days for children.

OR

- Erythromycin.

Management of cellulitis

❑ Pharmacologic

First line

- Procaine penicillin,
 - Children; 50,000u/kg/24 hrs. in a single dose for 10 days.
- If penicillinase resistant staphylococcus suspected:
 - Semi-synthetic penicillins (e.g. Cloxacillin 0.5 – 1gm every 4 hrs.) should be administered intravenously until the fever subsides, (usually 2 – 3 days). Then Cloxacillin 500 mg. P.O QID should be continued for 7 days.
 - Hospitalized patients should be treated with Crystalline Penicillin 1.2-2.4 million units IV 4 hourly.

Alternative

➤ Erythromycin.

❖ N.B. In all children with facial or periorbital cellulitis, coverage for *Haemophilus* *should be* provided with Chloramphenicol 30 to 50mg/kg/d divided in two four doses

OR

➤ Cephalexin, 500 mg P.O., QID.

Management of scabies

❑ Non pharmacologic

➤ washing clothes in hot water or ironing clothes after normal washing.

❑ Pharmacologic

➤ Permethrin 5% cream(first line) : applied to all areas of body from the neck down for 8-14 hrs then wash off. Repeat the same dose after a week.

➤ Benzyl benzoate – applied to the entire body, neck to toe for 3 to 5 consecutive evenings. Bath should be taken before the first and after the last application.

❖ ADE: Irritation of skin.

- Sulfur ointment- children 5%, thinly applied to the entire body for 3 consecutive nights.
- The patient should wash thoroughly before each new application and 24 hour after the last treatment.
- ❖ ADE: skin irritation
- ❖ Contraindications: pregnancy or lactation, children younger than 2 years.

Systemic

- Ivermectin (200 µg/kg) – as a single dose, for crusted scabies and resistant forms of scabies and it is ideal for institutional outbreaks.

Prevention

- ✓ Treat all households and close contacts simultaneously.
- ✓ Wash clothing, bedding, and towels used with in the previous 3 days in hot water (50oc) and seal them in plastic bag for at least 72 hours.
- ✓ Avoid close skin to skin contact until treatment is completed.

Prescription writing

Introduction

- Prescription is a written physician order choosing an appropriate medication or a device for a patient, in a given clinical situations.
- The prescription is the link between the prescriber, the pharmacist (or dispenser) and the patient so it is important for the successful management of the presenting medical condition

Content of a prescription

- A prescription should contain:
 - Name, address, age ,body weight of the medicine consumer and Date of the prescription;
 - Diagnosis; Generic name, dosage form and strength and directions for use of the medicines. The pharmaceutical form (for example ‘tablet’, ‘oral solution’, ‘eye ointment’) should also be stated.
 - prescriber’s name, signature and address.



Asella Referral & Teaching Hospital

PRESCRIPTION PAPER



Serial No. 319865

Name of Hospital: Asella Referral & Teaching Hospital

Patient's full name:

Tel: 251228126296

Sex:

Age:

Weight:

Card No.

Region:

Town:

Woreda

Kebele

House No:

Tel. No:

Outpatient

Inpatient

☐ Emergency

Diagnosis, if not NCD

[illegible]

Reconciliation: (Cash sales Ticket's Sr. NO and Sign. Cashier).

Reconciliation: (Cash sales, Ticket's SR No and Sig of dispenser) _____
Or (Registered code, SR No and Sig of dispenser for credit/free)

Prescriber's

Evaluato r's

Counselor's

Full name:

Qualification:

Registration:

Signature: _____

Date:

St. Paul's Hospital Millennium Medical College Printing Press

PRESCRIPTION ABBREVIATIONS

AAA	Apply to affected area	o.n.	Take every night
a.c.	Take before food	o.s.	Apply to left eye
a.d.	Apply to right ear	p.c.	Take after meals
ad lib.	Use as much as needed	p.o	Take by mouth (orally)
amp.	Ampoule	p.r.	Take rectally
a.s.	Apply to left ear	p.r.n.	Take as needed
a.u.	Apply to both ears	p.v.	Take vaginally
b.i.d.	Take two times daily	q12h	Take every 12 hours
bucc.	Apply to inside of cheek	q.a.m.	Take every morning
cap.	Capsule	q.d.	Take every day
DR	Delayed release	q.d.s	Take 4 times a day
EC	Enteric coated	q.o.d	Take every other day
ID	Intradermal	q.q.h	Take every 4 hours
IM	Intramuscular	SC	Subcutaneous
IN	Intranasal	Stat.	Immediately
Inj.	Injection	SC	Suppository
mane	Take in the morning	tab.	Tablet
neb.	Nebuliser	t.i.d.	Take 3 times a day
o.d.	Take once daily	top.	Apply topically
o.m.	Take every morning	tsp	Teaspoon

Abbreviations and Terms not to be used

- Ambiguous decimal points can cause 10-fold overdose.

example: write 0.1 mg, not .1 mg.

write 1 mg, not 1.0 mg (avoid trailing zero).

- Unsafe Abbreviations

Do not use U → write units.

Do not use µg/mcg → write microgram.

Do not use QD/Q.D. → write daily.

Do not use QOD → write every other day.

Do not use tsp/tbsp → write 5 mL/15 mL.

➤ Incomplete Prescriptions

Always specify the exact dose (mg), not just dosage form.

Example: Avoid "1 ampule of furosemide" (multiple strengths available).

➤ Write full drug names; avoid abbreviations (e.g., ASA, 6MP).

➤ Use generic names whenever possible.

➤ Mention the indication on the prescription (e.g., "for glaucoma").

- Example:- This is Naol Temesgen a 2 years old male child that developed a runny nose 5 days ago, followed by a cough and fever. Over the last 24 hours, his mother noticed rapid breathing, reduced appetite, and decreased activity. There is no history of choking or foreign body aspiration. Immunizations are up to date.

➤ Physical Examination:

- | | |
|--|---|
| • Weight: 10 kg | Respiratory rate: 42 breaths/min |
| • Oxygen saturation: 94% on room air | Temperature: 38.3°C |
| Heart rate: 120 beats/min | Mild chest indrawing |
| Crackles over the right lower lung field | Bronchial breath sounds in the right lower lobe |

➤ Investigations:

- Chest X-ray: Right lower lobe consolidation
- CBC: Elevated white blood cell count with neutrophil predominance

➤ Diagnosis: Community-acquired pneumonia (mild to moderate, suitable for outpatient management).

Amoxicillin oral suspension

- Dose: 30- 50 mg/kg/day TID for 5-7 days (total daily dose 150 mg/kg/day)
- For a 10 kg child: 500mg per dose, three times daily.
- If using a 250 mg/5 mL suspension, this is approximately 3.3~3mL PO every 8 hours for 5 days.

➤ Paracetamol oral suspension (160 mg/5 mL)

- Dose: 15 mg/kg/dose every 4–6 hours as needed for fever.
- For a 10 kg child: 150 mg (4.5mL) po PRN(as needed).
- Maximum: 4 doses in 24 hour

PRESCRIPTION PAPER

PRESSCRIPTION PATTER

Patient's full Name _____ Naoi Temelgen
Sex M Age 2 yrs Weight 10kg Card No. 781435
Region _____ Town Assela Woreda _____ Kebele _____
House No. _____ Tel. No. 09918222 In patient ☐ Out patient ☒
Specific Diagnosis, if not ICD Community acquired pneumonia (Cap)

Duration	Price
----------	-------

Drug Name Strength, Dosage Form, Dose, Frequency, Duration, Quantity, Route Of Administration AND Other Information	Price (dispensers use only)
R _x	
- Amoxi'cillin 250/5ml - 3ml po (suspension) TID for 5 days	
- para'cetamol 360 mg/5ml - 4.5 ml (suspension) po PRN	
Total Price	

Nur printing 0921-49-95-18

Nur printing 0921-49-95-18

**Reconciliation: (Cash sales Ticket's SR No. and Sig of cashier)
Or (Register code, SR No and Sig of dispenser for credit/free)**

Full Name Dr. Meron Yacob Evaluator's _____ Counselor's _____
Qualification GP _____
Registration # _____
Signature [Signature] _____
Date 15/11/2018 _____

Steps of rational prescribing

Step 1: Define the patient's problem

Step 2: Specify the therapeutic objective

Step 3: Verify the suitability of a drug

Step 4: Write a prescription

Step 5: Give information, instructions and warnings

Step 6: Monitor (and stop) the treatment

Which patients are most at risk of prescription error?

- patients on multiple medications
- patients with another condition, e.g. renal impairment, pregnancy
- patients who cannot communicate well
- patients who have more than one doctor (**medicine reconciliation**)
- patients who do not take an active role in their own medication use
- children and babies (dose calculations required)

What you can do to make medication use safer

- Know the medication you prescribe well
- Use memory aids
- Develop checking habits

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